

I. Relativistic Kinematic

1. A spaceship with a rest length of 100 m passes in front of an Earth observer in $4\mu\text{s}$. What is its velocity relative to the Earth?
2. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 20 ns; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is 7.5 ns . What is the propagation velocity of the particle beam?
3. A certain bacterium doubles its population in 20 days. Two such bacteria are placed in a spaceship and sent on a 1000-day space journey. During the trip, the spaceship moves at a constant speed of $0.995c$. How many bacteria will be on board at the moment of return?
4. Aladár drives a nail into point $P_1(4000\text{ m}, -100\text{ m}, 200\text{ m})$ in the coordinate system K . At that moment, the clock in K shows $t = 15\mu\text{s}$. At the same time in K , Balázs drives another nail into point $P_2(3000\text{ m}, 500\text{ m}, 150\text{ m})$ of the same coordinate system. (The two events occur simultaneously in K .) Calculate the time interval between the two events in a reference frame K' moving at a uniform velocity $v = 2 \cdot 10^8\text{ m/s}$, parallel to the x-axis.

II. Relativistic Dynamics

- (a) A particle moves with a velocity such that $\frac{v}{c} = 0.99$. Determine the value of $\frac{m}{m_0}$!
- (b) Determine the relativistic mass and the velocity of an electron whose kinetic energy is 100 keV ($1.6 \cdot 10^{-14}\text{ J}$). The rest mass of the electron is $9.11 \cdot 10^{-31}\text{ kg}$.
- (c) Calculate the rest energy of an electron, corresponding to its rest mass of $9.11 \cdot 10^{-31}\text{ kg}$, in joules and in electronvolts! ($1\text{ MeV} = 1.6 \cdot 10^{-13}\text{ J}$)

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$$\begin{array}{c}
 \xrightarrow{v} t' \\
 \hline
 t' = ? \quad N_0 = ? \\
 \downarrow \\
 N = ?
 \end{array}$$

3. A certain bacterium doubles its population in 20 days. Two such bacteria are placed in a spaceship and sent on a 1000-day space journey. During the trip, the spaceship moves at a constant speed of 0.995 c. How many bacteria will be on board at the moment of return?

$$t = t' / \sqrt{1 - v^2/c^2}$$

$$\begin{aligned}
 t' &= t \cdot \sqrt{1 - v^2/c^2} = 1000 \text{ days} \cdot \sqrt{1 - (0.995^2 \cdot c^2)/c^2} = 1000 \text{ days} \cdot \sqrt{1 - 0.995^2} = ? \\
 &= 1000 \text{ day} \cdot 0.0999 \sim 1000 \text{ day} \cdot 0.1 = 100 \text{ days}
 \end{aligned}$$

t' : in moving reference coordinate system

t : our reference system

$$N = N_0 \cdot 2^{(100 \text{ days} / 20 \text{ days})} = 2 \cdot 2^5 = 2^6 = 256$$

$(Q) \rightarrow m$
 $\frac{v}{m} = 0.99c$
 $m_0 = ?$
 $(h) W_{kin} \rightarrow e^-$
 $9.11 \cdot 10^{-31} \text{ kg}$
 $m = ?$
 $v = ?$

II. Relativistic Dynamics
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(a) $m = m_0 / \sqrt{1 - v^2/c^2}$

$m/m_0 = 1 / \sqrt{1 - v^2/c^2} = 1 / \sqrt{1 - 0.99^2} = 7.09$

(b) $W_{kin} = m_0 c^2 / \sqrt{1 - v^2/c^2} - m_0 c^2$

$W_{kin} = m_0 c^2 (1 / \sqrt{1 - v^2/c^2} - 1)$
 $W_{kin} / (m_0 c^2) + 1 = 1 / \sqrt{1 - v^2/c^2}$
 $\sqrt{1 - v^2/c^2} = 1 / (1 + W_{kin} / (m_0 c^2))$
 $1 - v^2/c^2 = 1 / (1 + W_{kin} / (m_0 c^2))^2$
 $v = c \cdot \sqrt{1 - 1 / (1 + W_{kin} / (m_0 c^2))^2}$

$W_k = m_0 c^2 / \sqrt{1 - v^2/c^2}$

$- m_0 c^2 =$
 $= m_0 c^2 \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)$

$\frac{W_k}{m_0 c^2} = \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right)$

$1 + \frac{W_k}{m_0 c^2} = \frac{1}{\sqrt{1 - v^2/c^2}}$
 $\sqrt{1 - v^2/c^2} = \frac{1}{1 + \frac{W_k}{m_0 c^2}}$

$1 - \frac{v^2}{c^2} = \left(\frac{1}{1 + \frac{W_k}{m_0 c^2}} \right)^2$
 $\frac{v^2}{c^2} = 1 - \left(\frac{1}{1 + \frac{W_k}{m_0 c^2}} \right)^2$

$V^2 = c^2 \cdot \left(1 - \left(\frac{1}{1 + \frac{W_k}{m_0 c^2}} \right)^2 \right)$
 $V^2 = c^2 \cdot \left(1 - \left(\frac{1}{\frac{W_k + m_0 c^2}{m_0 c^2}} \right)^2 \right)$

$V^2 = c^2 \cdot \left(1 - \left(\frac{m_0 c^2}{W_k + m_0 c^2} \right)^2 \right)$
 $V = c \cdot \sqrt{1 - \left(\frac{m_0 c^2}{W_k + m_0 c^2} \right)^2}$

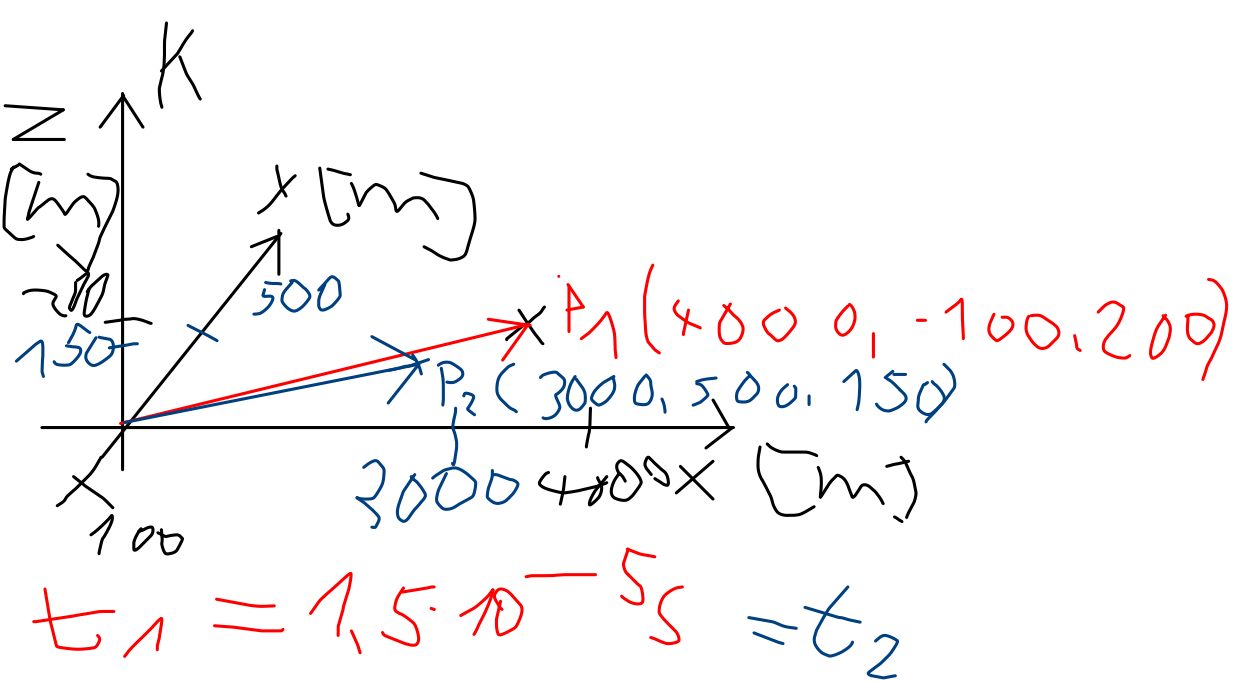
$v = 3 \cdot 10^8 \text{ m/s} \cdot \sqrt{1 - \left(\frac{1}{1 + 1.6 \cdot 10^{-14} \text{ J} / (9.11 \cdot 10^{-31} \text{ kg} \cdot (3 \cdot 10^8 \text{ m/s})^2)} \right)^2} = 3 \cdot 10^8 \text{ m/s} \cdot \sqrt{1 - (1 / (1 + 0.1951))^2}$
 164275500.07 m/s

$$v=c*\sqrt{1-1/(1+W_{\text{kin}}/(m_0*c^2))^2}$$

$$m=m_0/\sqrt{1-v^2/c^2}=9.11*10^{-31} \text{ kg}/\sqrt{1-(164275500,07 \text{ m/s}/(3*10^8))^2}=9,11*10^{-31} \text{ kg}*1.43=1.3*10^{-30} \text{ kg}$$

(c) Calculate the rest energy of an electron, corresponding to its rest mass of $9.11*10^{-31} \text{ kg}$, in joules and in electronvolts!

$$E=m*c^2=9.11*10^{-31} \text{ kg} * (3*10^8 \text{ m/s})^2=8.199*10^{-14} \text{ J}$$



$$\Delta t = ?$$

$$L = 2 \cdot 10^8 \frac{\text{m}}{\text{s}} \parallel x\text{-axis}$$

So, the events occurred in the same time in the same reference system. The time difference is zero and it will remain zero in any coordinate system. Therefore, $dt=0$ s.